ODSC AI

THE #1 AI TRAINING CONFERENCE

AI AGENTS

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SAN FRANCISCO | OCT 28-30

INTRO TO AGENTS USING SMOLAGENTS

Intro to AI Agents using smolagents

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ODSC AI WEST 2025

Learning Objectives and Tools

- Understand the fundamental differences between traditional LLMs and AI agents.
- Explain the concept of Levels of Agency and when to apply each
- Apply the ReAct approach (Reasoning + Acting) to structure agent workflows
- Build AI agents using smolagents, integrating tools to extend capabilities
- Create custom tools and connect them with agents for practical use cases
- Create more complex advanced use cases.

Learning Objectives and Tools

- smolagents
- Google Colab (notebooks)
- Hugging Face account (to use a token)
  - Better if we're PRO

Hi, I'm Sergio 👋

- **Machine Learning Engineer @ Hugging Face**
  - Multimodality, Post-Training, Agents...
- **PhD in AI**
- **Open source and AI → GSoC / ML Circle Madrid**
- **Teaching and Sharing**

Sergio Paniego Blanco (@sergiopaniego)

What is an AI Agent?

System that leverages an AI model to interact with its environment in order to achieve a user-defined objective.
It combines reasoning, planning, and the execution of actions (often via external tools) to fulfill tasks

AI components

- Brain (AI model) 🧠
- Body (capabilities and tools) 🔧

AI models: LLM (text-to-text) or VLM (image-text-to-text).

- GPT 5 from OpenAI, Qwen models, Llama from Meta, Gemini from Google …

How do they take actions? → they have access to tools!

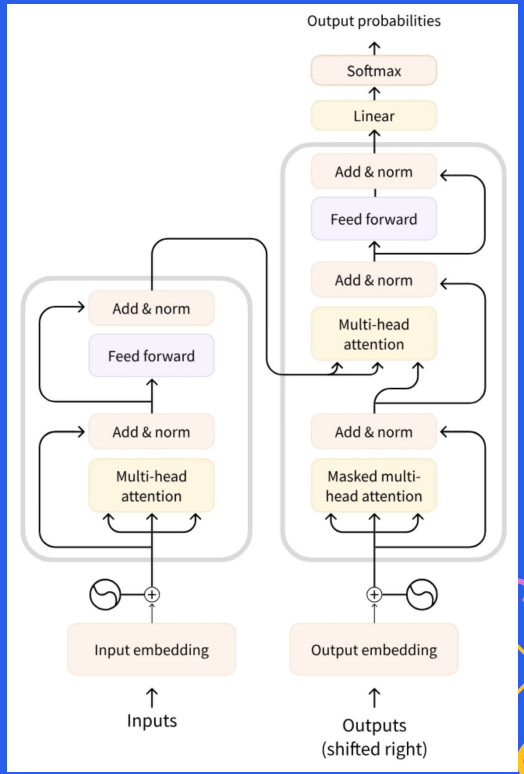
Levels of Agency

Agency level	Description	Short name	Example code
	LLM output has no impact on program flow	Simple processor	process_llm_output(llm_response)
★	LLM output controls an if/else switch	Router	If llm_decision(): path_a() else: path_b()
★★	LLM output controls function execution	Tool call	run_function(llm_chosen_tool, llm_chosen_args)
★★	LLM output controls iteration and program continuation	Multi-step Agent	while llm_should_continue(): execute_next_step()
★★★	One agentic workflow can start another agentic workflow	Multi-Agent	If llm_trigger(): execute_agent()
★★★	LLM acts in code, can define its own tools/start other agents	Code Agents	def custom_tool(args): ...

What is an LLM?

AI system that excels at understanding and generating human language.

- Trained on vast amounts of text data (multimodal too)
- They use transformers 🤖
  - Encoders
  - **Decoders**
  - Seq2Seq (encoder-decoder)



Types of transformers

Encoders:

- Example: ModernBERT, BERT
- Use case: text classification, semantic search, NER
- Size: millions of params

Decoders:

- Example: Llama, Gemini, GPT…
- Use case: text generation, chatbots, code generation.
- Size: billions of params

Seq2Seq (encoder-decoder):

- Example: T5, BART
- Use case: translation, summarization
- Size: millions of params

Typical LLM systems are decoders: DeepSeek, GPT, Gemini, Llama, Gemma, SmolLM…

Goal of LLMs

Predict the next token given a sequence of previous tokens (autoregressive)

Base vs Instruct models

Base model: trained on raw data to predict the next token
Instruct model: fine-tuned to follow instructions and engage in conversations. Use chat templates

```
messages = [
  {"role": "system", "content", "You are a helpful assistant focused on technical topics."},
  {"role": "user", "content", "Can you explain what a chat template is?"},
  {"role": "assistant", "content", "A chat template structures conversations between users and AI models..."},
  {"role": "user", "content", "How do I use it?"},
]
```

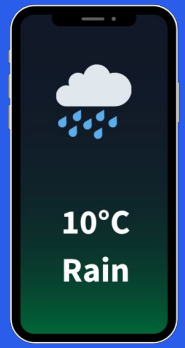
What are tools?

Function given to the LLM with a clear goal

- Web search, image generation, retrieval, API interface...

What's the current weather in Paris?

The LLM hallucinates



Agent that has WebSearch Tool

What are tools?

Function given to the LLM with a clear goal

- Web search, image generation, retrieval, API interface...

```
system_message="""You are an AI assistant designed to help users efficiently an accurately. Your primary goal is
to provide helpful, precise, and clear responses.

You have access to the following tools:
Tool Name: calculator, Description: Multiply two integers., Arguments: a: int, b: int, Outputs: int
"""
```

AI Agent workflow

Thinking (Thought) → Acting (Act) and observing (Observe)

AI Agent workflow

Thinking (Thought) → Acting (Act) and observing (Observe)

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system_message="""You are an AI assistant designed to help users efficiently an accurately. Your primary goal is
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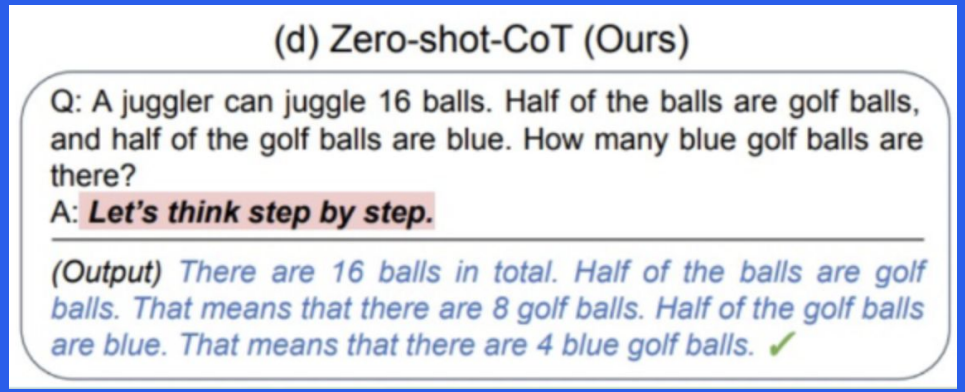
You have access to the following tools:
Tool Name: calculator, Description: Multiply two integers., Arguments: a: int, b: int, Outputs: int

You should think step by step in order to fulfill the objective with a reasoning divided into
Thought/Action/Observation steps that can be repeated multiple times if needed.

You should first reflect on the current situation using `Thought: {your_thoughts`}, then (if necessary), call a
tool with the proper JSON formatting `Action: {JSON_BLOB}`, or print your final answer starting with the prefix
`Final Answer:`
"""
```

Thought: Internal Reasoning & ReAct

- Thoughts represent the Agent's internal reasoning and planning processes to solve the task
- **ReAct approach:** "Reasoning" (Think) + "Acting" (Act). Prompting technique that appends "Let's think step by step" before letting the LLM decode the next tokens.

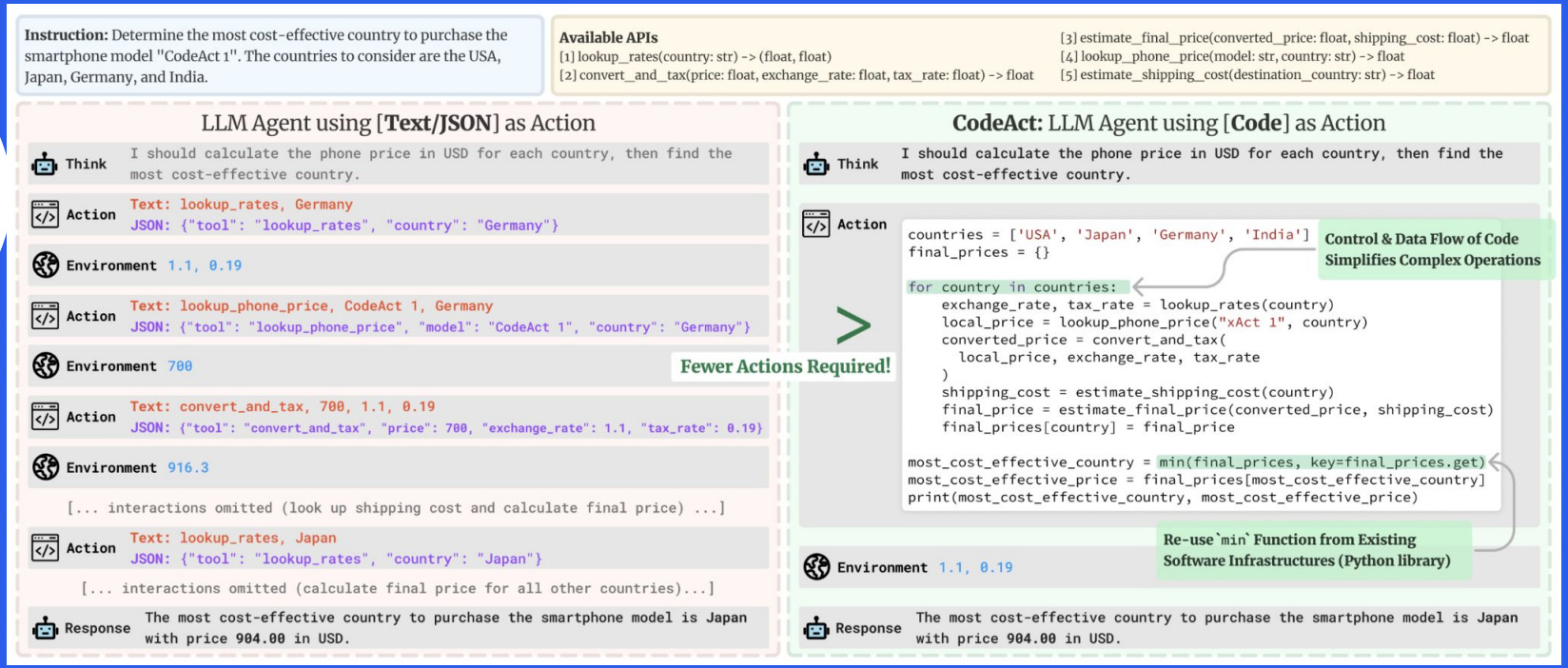


Actions: Engaging with environment

- Actions are concrete steps an AI agent takes to interact with its environment.
- Types of Agent Actions:
  - **Code Agent:** agent writes code block that is interpreted externally.
  - **JSON Agent:** action is specified in JSON format.

```
Thought: I need to check the current weather for New York
Action:
{
  "action": get_weather",
  "action_input": {"location", "New York"}
}
```

Actions: Engaging with environment



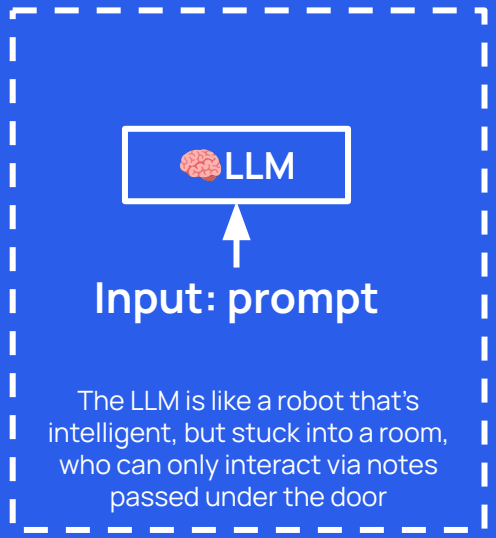
Observe: Integrating Feedback to Reflect and Adapt

- How an agent perceives the consequences of its actions.
- Observation phase:
  - Agent collects feedback
  - Appends results
  - Adapts its strategy
- How are results appended?
  - Parse the action
  - Execute the action
  - Append the result as an observation

ReAct agent

Why an agent?

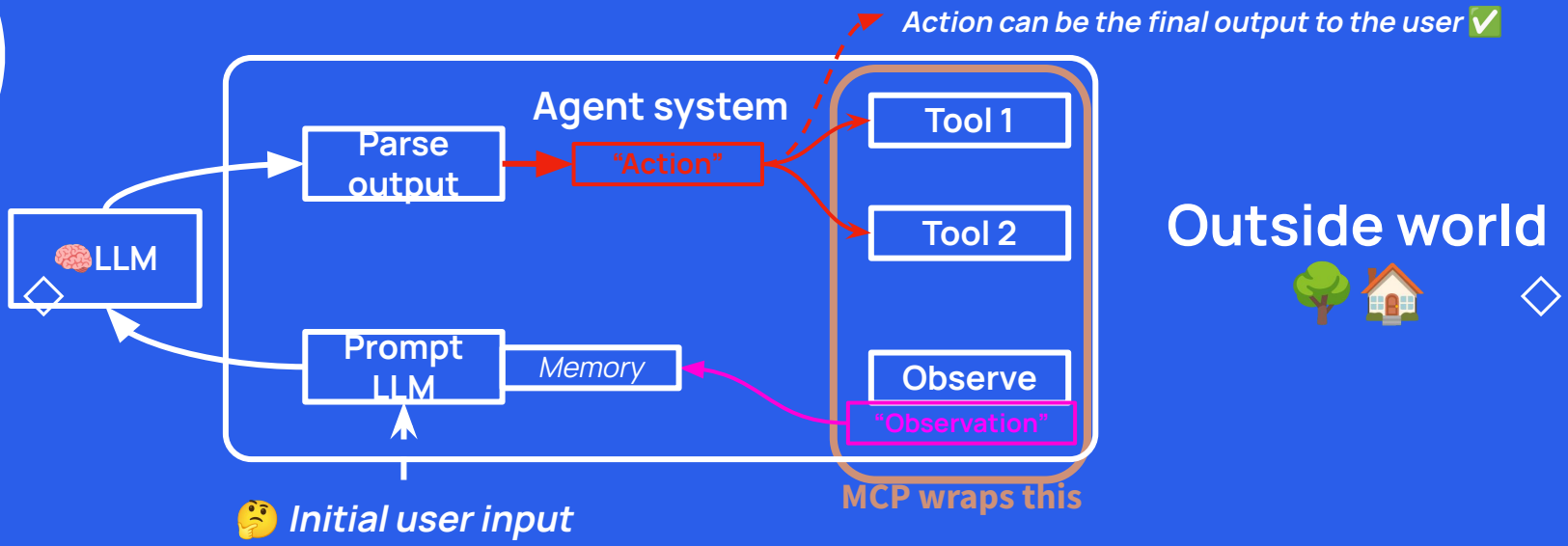
Need to de-isolate an LLM from the world.



Outside world

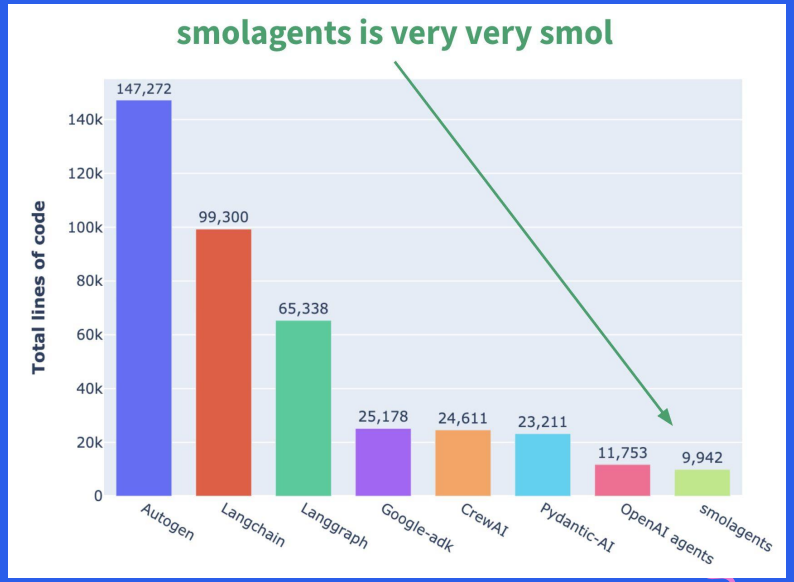
What is MCP?

Protocol that proposes and easy wrapper on tools (and prompts)



Agentic Frameworks

smolagents (Hugging Face)
LangGraph (LangChain)
Llama-Index
OpenAI Agents SDK (OpenAI)
Autogen by Microsoft



smolagents

LlamaIndex

LangGraph

AG

Why smolagents

- **smolagents** is a simple yet powerful framework for building AI agents.
- **Advantages:**
  - Simplicity
  - Flexible LLM Support (transformers, HfApiModel, vLLM, OpenAI...)
  - Code-First Approach
  - HF Hub integration

Why smolagents

In smolagents, agents operate as multi-step agents:

- One thought
- One tool call and execution

Type of agents:

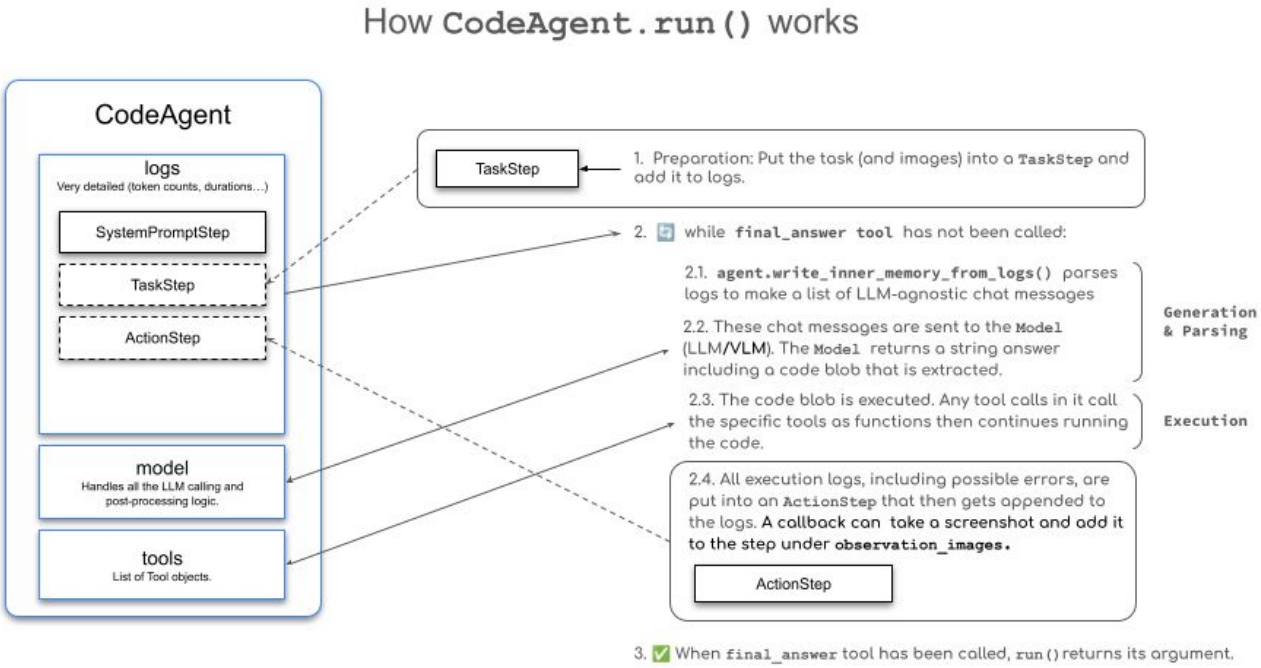
- CodeAgent! 👩‍💻
- ToolCallingAgent 🔧

Building Agents that use code

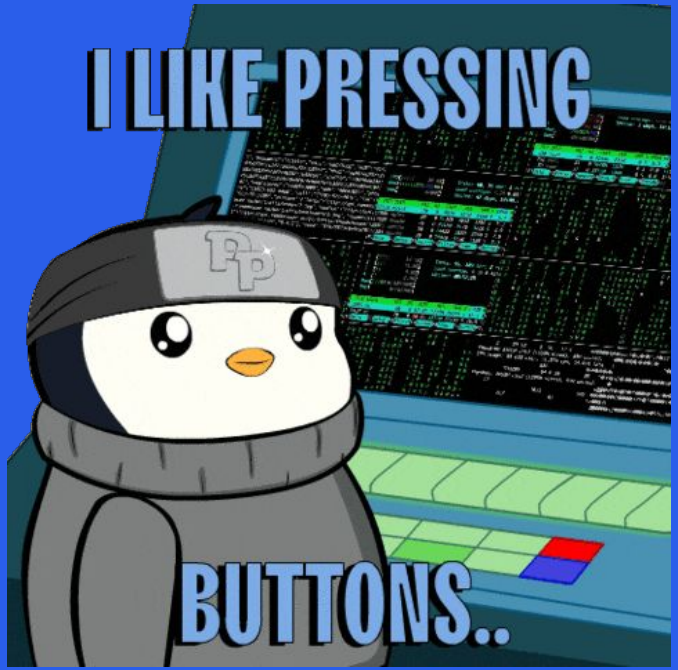
Why Code Agents?

- **Composability:** easily combine and reuse actions
- **Object Management:** work directly with complex structures like images
- **Generality:** express any computationally possible task
- **Natural for LLMs:** High-quality code is already present in LLM training data

Building Agents that use code



Building Agents that use code



Notebook

Writing actions as code snippets

Use built-in tool-calling capabilities of LLM providers to generate tool calls as JSON structures

```
for query in [
    "Best catering services in Gotham City",
    "Party theme ideas for superheroes"
]:
    print(web_search(f"Search for:{query}"))
```

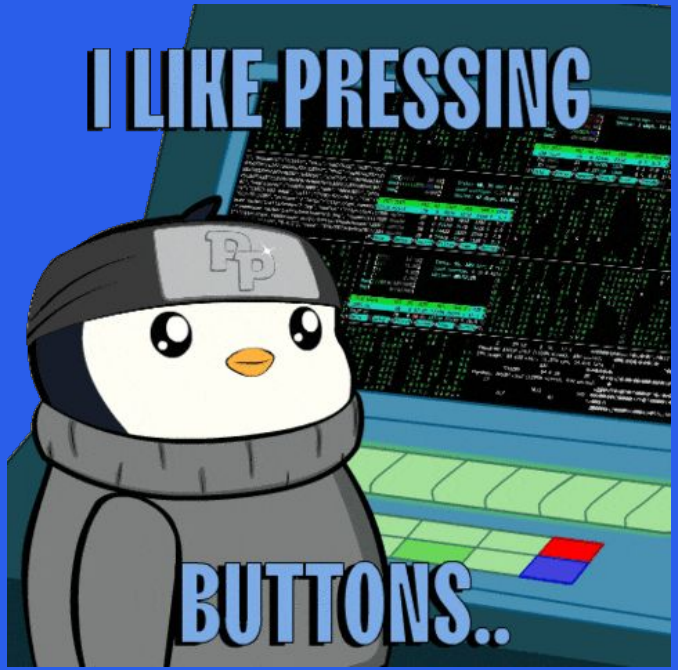
Tools

- The LLM need an interface description with:
  - **Name:** web_search
  - **Tool description:** searches the web for specific queries
  - **Input types and descriptions:** query (string)
  - **Output types:** string containing the search results.
- How are they created:
  - @tool decorator
  - Subclass of Tool

Tools

- smolagents also has a default toolbox:
  - PythonInterpreterTool
  - FinalAnswerTool
  - UserInputTool
  - DuckDuckGoSearchTool
  - GoogleSearchTool
  - VisitWebpageTool
- Tools can be shared and imported:
  - To/from HF Hub
  - From HF Spaces
  - From LangChain Tools
  - From MCP Servers

Tools

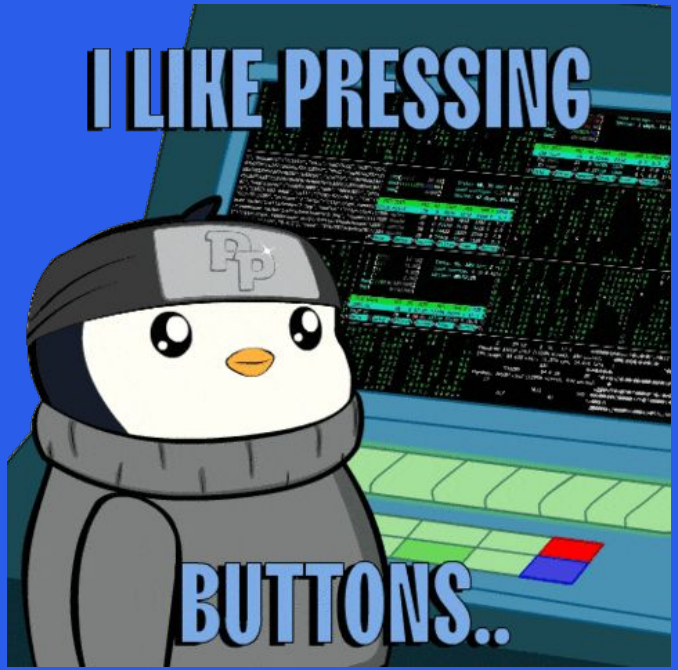


Notebook

(Bonus) Retrieval Agents

- Retrieval Augmented Generation (RAG) systems combine the capabilities of data retrieval and generation models to provide context-aware responses.
- **Agentic RAG** → combines autonomous agents with dynamic knowledge retrieval.
- We could retrieve from:
  - The web
  - Custom knowledge bases using a specific tool
  - We could enhance these systems with more capabilities

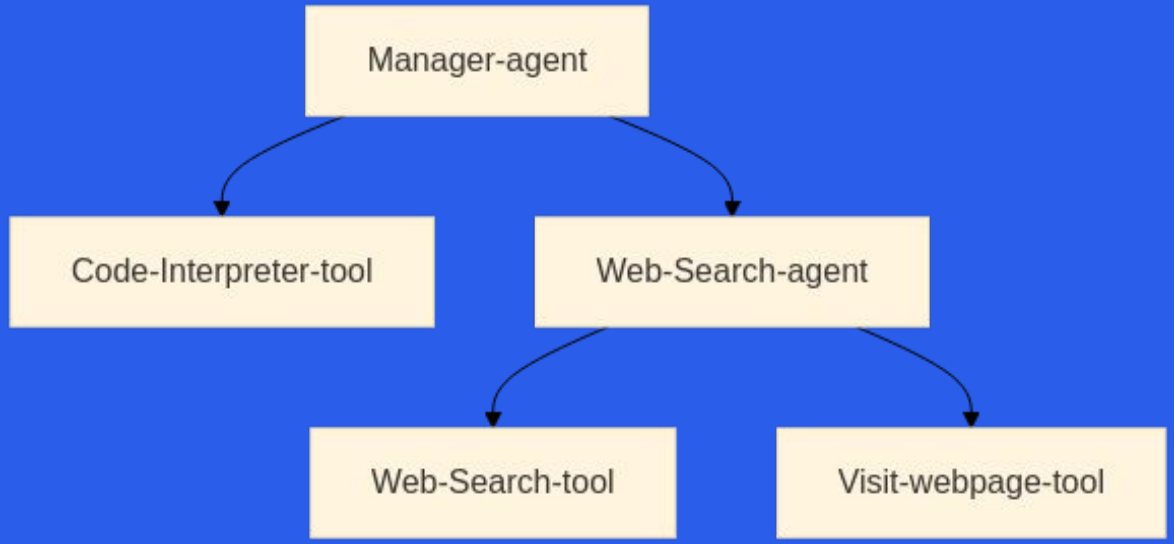
(Bonus) Retrieval Agents



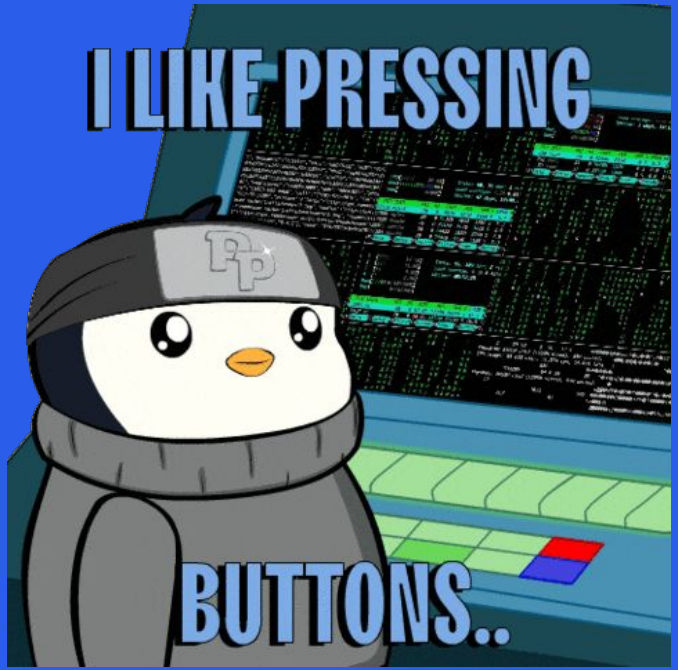
Notebook

(Bonus) Multi-Agent Systems

Enable specialized agents to collaborate on complex tasks.



(Bonus) Multi-Agent Systems



Notebook

If you want to know more

Agents Course

Thanks!

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